
GROWING DIFFERENTLY: FINANCIAL CYCLES, AUSTERITY, AND COMPETITIVENESS IN GROWTH MODELS SINCE THE GLOBAL FINANCIAL CRISIS*

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ABSTRACT

COVID-19 triggered the largest peacetime fiscal expansion in history. Did this represent a structural shift toward state-led growth, or a temporary crisis response? We estimate government contribution to GDP growth for 21 OECD economies (1995–2023) using country fixed effects with period dummies. The COVID-era government contribution increased by 0.25 percentage points relative to the post-GFC austerity trough ($p = 0.003$), far below the hypothesized threshold for structural transformation. When 2020 is excluded, the effect collapses to 0.13 points and loses significance ($p = 0.407$). The strongest and most robust finding is financial-cycle amplification: a 10-percentage-point higher pre-pandemic credit-to-GDP gap predicts a 0.17-point larger government contribution increase during COVID ($p < 0.001$). Pre-pandemic growth model type explains little beyond financial structure. We find no evidence for a durable state-led transformation. The fiscal expansion was a crisis response concentrated in 2020, not a regime shift.

Keywords growth models; financial cycles; fiscal policy; COVID-19; state-led growth; credit-to-GDP gap; austerity; comparative political economy

1 Introduction

Governments spent roughly \$16 trillion fighting the COVID-19 recession—direct transfers to households, wage subsidies, equity injections, and loan guarantees deployed at a speed and scale with no peacetime precedent. Observers from across the political spectrum asked the same question: does this mark a permanent turn toward state-led growth, or a temporary crisis response that will reverse once the pandemic recedes? Stiglitz (2021) argued for a paradigm shift. IMF Fiscal Monitor (2021) documented the scale. But no systematic empirical assessment has tested whether the government’s role in driving aggregate demand growth actually changed in a durable way.

We answer this question using a balanced panel of 21 OECD economies spanning 1995 to 2023. We estimate the government contribution to real GDP growth—the combined contribution of government consumption and public investment—using country fixed effects with period dummies. This design isolates within-country changes across three macroeconomic regimes: the pre-GFC financialization era (1995–2007), the post-GFC austerity era (2010–2019), and the COVID-era fiscal expansion (2020–2023). We test whether the COVID-era government contribution exceeds the hypothesized threshold of 1.0 percentage points of GDP growth, persists beyond 2020, and varies with pre-pandemic growth model type and financial cycle position.

We find no evidence of a durable state-led transformation. The government contribution to growth increased by only 0.25 percentage points during COVID relative to the post-GFC baseline ($p = 0.003$)—a quarter of the hypothesized threshold—and this increase was driven entirely by 2020. When we exclude the acute pandemic year and re-estimate for 2021–2023, the effect drops to 0.13 points and loses statistical significance ($p = 0.407$). The COVID-era government contribution (0.48 percentage points) does not exceed pre-GFC levels (0.66 points), confirming that the fiscal expansion represents a partial recovery from post-GFC austerity rather than a structural regime shift. Our strongest

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finding is financial-cycle amplification: countries with elevated pre-pandemic credit-to-GDP gaps deployed substantially more government spending during COVID—a 10-percentage-point higher gap predicts an additional 0.17-point increase ($p < 0.001$). This result is robust across all specifications and directly addresses Nölle (2023)’s observation that financial cycle dynamics remain under-theorized in the growth models framework.

We contribute to three literatures. First, we provide the first empirical test of the state-led growth model hypothesis in the comparative political economy tradition, extending the growth models taxonomy (Baccaro and Pontusson, 2016, 2022) into the COVID era and finding that the framework’s core categories are weaker predictors of government spending patterns than financial structure. Second, we integrate financial cycle dynamics—measured via the Borio and Drehmann (2009) credit-to-GDP gap—into growth model operationalization, showing that credit-cycle position shapes crisis fiscal response more than the export-led/consumption-led/debt-led taxonomy would predict. Third, we add to the post-GFC fiscal policy literature (Blanchard and Leigh, 2013; Fatás and Summers, 2018) by showing that the extraordinary COVID-era fiscal expansion did not produce a hysteresis-like ratchet effect on the government’s growth role. The paper proceeds as follows: Section 2 situates our contribution within the growth models, financial cycles, and fiscal policy literatures. Section 3 describes the data and sample construction. Section 4 presents the estimation strategy. Section 5 reports results. Section 6 concludes.

2 Related Literature

Our paper sits at the intersection of three research programs: the growth models approach in comparative political economy, the macroeconomics of financial cycles, and the post-GFC fiscal policy literature.

The growth models framework, launched by Baccaro and Pontusson (2016, 2022), argues that the Varieties of Capitalism distinction between liberal and coordinated market economies (Hall and Soskice, 2001) fails to capture post-GFC growth dynamics. Instead, growth is sustained by distinct demand regimes—export-led, consumption-led, or debt-led—each stabilized by political coalitions that defend the regime’s distributive implications. Baccaro and Hadziabdic (2022) provide the canonical operationalization through demand decomposition of GDP growth by expenditure component. Hope and Soskice (2016) link growth models to macroeconomic dynamics and political institutions. Hassel and Palier (2021) trace how growth regimes and welfare states co-evolved from the post-war period through the post-GFC era. Nölle (2023) surveys the methodological landscape and identifies two critical gaps: the framework lacks integration with financial cycle dynamics, and no empirical study covers the post-2019 period. Our paper directly addresses both gaps. We extend the demand decomposition methodology to 2023 and incorporate the Borio and Drehmann (2009) credit-to-GDP gap as a financial-cycle variable. Our finding that financial cycle position outperforms growth model type in predicting COVID-era government spending heterogeneity qualifies a core premise of the framework—that the export-led/consumption-led/debt-led taxonomy captures the most relevant structural variation.

A parallel literature documents the macroeconomics of financial cycles. Borio (2014) establishes that financial cycles are 16–20-year phenomena distinct from business cycles, with credit growth as their defining feature. Schularick and Taylor (2012) show that credit expansion is the best predictor of financial crises. Jordà, Schularick, and Taylor (2016) demonstrate that mortgage lending has displaced business lending as the primary driver of financial instability. Stockhammer (2015) and Kumhof, Rancière, and Winant (2015) link rising inequality to debt-financed consumption and financial fragility. Stockhammer, Wildauer, and Kohler (2020) provide the bridge to growth models: they show that household debt contributed positively to growth before 2008 but the relationship turned negligible or negative after the GFC, suggesting that debt-financed growth has a finite lifespan. Our finding that high-credit-gap economies deployed more government spending during COVID extends this logic: when private credit channels are exhausted, government spending becomes the residual growth driver, but only during the acute crisis.

The third literature concerns fiscal policy and the post-GFC reassessment of austerity. Blanchard and Leigh (2013) demonstrated that fiscal multipliers during the post-GFC period were roughly three times larger than policymakers assumed (approximately 1.5 versus 0.5), directly undermining the intellectual foundation of the expansionary austerity hypothesis (Alèsina and Ardagna, 2010). Auerbach and Gorodnichenko (2012) showed that multipliers are state-dependent, reaching 2–3 in recessions. DeLong and Summers (2012) argued that fiscal expansion can be self-financing in depressed economies. Fatás and Summers (2018) documented permanent output losses from fiscal consolidation through hysteresis channels. The COVID-19 pandemic appeared to complete this intellectual transformation: governments deployed fiscal expansion at historically unprecedented scale without triggering the sovereign debt crises that austerity advocates had predicted (Gürkaynak and Wright, 2022). However, our findings suggest that this transformation did not extend to a durable restructuring of the government’s growth role. The COVID-era fiscal expansion was large but temporary—the hysteresis logic that Fatás and Summers (2018) identified in the austerity direction did not operate in reverse.

Mazzucato (2013) and Block (2008) provide the theoretical foundation for the state-led growth model concept, documenting the state’s central role in innovation even in liberal market economies. Rodrik (2004) reframes industrial policy as strategic collaboration rather than picking winners. Stiglitz (2021) argued that the COVID-19 fiscal response constitutes a paradigm shift toward a more active state. Our paper operationalizes this concept within the growth models framework and provides the first systematic empirical test. We find no evidence that the paradigm shift argument generalizes to the aggregate growth contribution of the state across advanced OECD economies.

3 Data

We construct a balanced country-year panel of 21 OECD advanced economies covering 1995–2023 (609 observations). The core sample includes Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Ireland, Italy, Japan, the Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, Switzerland, the United Kingdom, and the United States.

3.1 Dependent Variable

Our dependent variable is the government contribution to real GDP growth, measured in percentage points. We compute this as the sum of two components: the real growth rate of government final consumption expenditure weighted by its lagged nominal GDP share, plus the real growth rate of government gross fixed capital formation weighted by its lagged share. This follows the demand decomposition methodology of Baccaro and Hadziabdic (2022), adapted to isolate the government component. Across the full sample, the government contribution averages 0.49 percentage points ($SD = 0.66$), declining from 0.66 points pre-GFC (1995–2007) to 0.23 points post-GFC (2010–2019), before partially recovering to 0.48 points during the COVID era (2020–2023).

3.2 Growth Model Classification

We classify each country’s pre-pandemic growth model type using the average demand decomposition over 2010–2019, following Baccaro and Hadziabdic (2022). A country is export-led if net exports are the largest positive contributor to growth; consumption-led if household consumption dominates; debt-led if consumption and investment contributions together exceed net exports and household debt-to-GDP is elevated. Nine countries are classified as export-led (Austria, Denmark, Finland, Germany, Ireland, the Netherlands, Norway, Sweden, Switzerland), six as consumption-led (Australia, Belgium, Canada, France, Japan, New Zealand), and six as debt-led (Greece, Italy, Portugal, Spain, the United Kingdom, the United States).

3.3 Financial Cycle Measure

Our financial cycle measure is the credit-to-GDP gap—the deviation of private non-financial sector credit as a percentage of GDP from its long-run trend, estimated via a one-sided HP filter with smoothing parameter $\lambda = 400,000$, following Borio and Drehmann (2009). Data are from the BIS credit statistics, annualized to Q4. The gap has a mean near zero by construction with a standard deviation of 19.3 percentage points and substantial cross-country variation at the onset of the pandemic, ranging from deeply negative (credit contraction) in deleveraging economies to strongly positive (credit booms) in others.

3.4 Sources and Controls

GDP and expenditure component data come primarily from the World Bank’s World Development Indicators and the OECD National Accounts. Control variables include log GDP per capita, trade openness, population growth, the unemployment rate, public debt-to-GDP, the central bank policy rate, the real effective exchange rate, household debt-to-GDP, and the old-age dependency ratio. Data sources are mixed: 17 indicators are sourced directly from the World Bank API at coverage exceeding 90%; BIS credit series are obtained via the FRED API. Government GFCF, household debt-to-GDP, REER, and policy rate series involve literature-calibrated synthetic values where direct measurement was unavailable, a limitation we discuss below. Missing values within series are handled via linear interpolation for gaps of up to five years. The final estimation sample contains 563 complete observations after dropping 46 country-years (7.6%) with missing covariates.

3.5 Limitations

Three data limitations warrant mention. First, government gross fixed capital formation is estimated from literature-calibrated parameters rather than sourced directly from national accounts, introducing measurement error in the government contribution variable. Second, the credit-to-GDP gap is subject to known endpoint instability in the HP filter, making 2020–2023 gap values less reliable than those for earlier periods. Third, 2020 GDP reflects simultaneous supply and demand shocks, weakening the standard assumption that expenditure-side growth decomposition identifies demand-side contributions alone.

4 Methodology

4.1 Identification Strategy

We estimate the government contribution to GDP growth using country fixed effects with period dummies. The identification challenge is that COVID-19 was a simultaneous global shock—all advanced economies deployed extraordinary fiscal expansion, eliminating any untreated control group for a difference-in-differences design. Our strategy instead exploits within-country variation across three structurally distinct macroeconomic regimes: pre-GFC (1995–2007), post-GFC (2010–2019), and the COVID era (2020–2023). Country fixed effects absorb all time-invariant heterogeneity—institutions, geography, political culture, legal origin. The coefficients on period dummies identify the average change in government contribution relative to the post-GFC reference period, net of country-specific means.

Our baseline specification is:

$$G_contrib_{it} = \alpha + \beta_1 \cdot \text{PreGFC}_t + \beta_2 \cdot \text{GFC}_t + \beta_3 \cdot \text{COVID}_t + \gamma \cdot \text{Controls}_{it} + \mu_i + \varepsilon_{it} \quad (1)$$

where $G_contrib$ is the government contribution to real GDP growth in percentage points, PreGFC, GFC, and COVID are period dummies (reference = post-GFC 2010–2019), Controls includes nine macroeconomic covariates, and μ_i are country fixed effects. The coefficient β_3 is our central parameter of interest—the average COVID-era shift. We build on this baseline with three extensions: Model 2 adds COVID $\times$ growth model type interactions (export-led as reference); Model 3 adds the credit-to-GDP gap and its interaction with the COVID dummy; Model 4 includes all interactions jointly. Standard errors are clustered at the country level to account for within-country serial correlation (Bertrand, Duflo, and Mullainathan, 2004).

4.2 Threats to Validity

The central identification threat is that the COVID-era government contribution increase may reflect automatic stabilizers and GDP denominator effects rather than discretionary fiscal expansion. The increase in government’s GDP share during 2020 partly reflects the mechanical effect of a collapsing private-sector denominator, not a deliberate growth strategy. We address this through two strategies. First, we re-estimate the COVID effect for 2021–2023 only, excluding 2020, to isolate the post-acute-crisis period when GDP had partially recovered and automatic stabilizer effects had receded. Second, we control for unemployment (the primary automatic stabilizer channel) and GDP per capita in all specifications.

A second threat is that fiscal response magnitude was endogenous to pandemic severity. Countries hit harder by COVID—measured by excess mortality or lockdown stringency—deployed larger fiscal responses for reasons unrelated to growth model transformation. Our fixed effects absorb time-invariant pandemic vulnerability (such as population density or healthcare capacity), and our COVID $\times$ credit gap interaction specification exploits pre-pandemic financial structure—which is predetermined with respect to pandemic severity—as the primary source of identifying variation in fiscal response heterogeneity.

The short COVID-era observation window (four years, 2020–2023) limits our ability to distinguish temporary from permanent changes. We frame our durability test conservatively: we ask whether the government contribution increase is statistically distinguishable from zero for 2021–2023, not whether it is “permanent.” A definitive assessment will require data through at least 2025–2027.

4.3 Robustness Checks

We conduct four robustness checks. R1 redefines the COVID era as 2021–2023, excluding 2020—our primary durability test. R2 uses pre-GFC rather than post-GFC as the reference period. R3 excludes the six debt-led economies to

test whether results are driven by Southern European sovereign debt crisis dynamics. R4 adds a linear time trend to control for secular changes in government size unrelated to period-specific shocks.

5 Results

5.1 The COVID-Era Government Contribution

The COVID-era government contribution to GDP growth increased by 0.25 percentage points relative to the post-GFC baseline ($p = 0.003$, 95% CI: 0.09 to 0.42). The effect is statistically significant but economically modest—0.38 standard deviations of the dependent variable and only one-quarter of the 1.0-point threshold we hypothesized for a structural regime shift. The pre-GFC period also shows a positive coefficient (0.22 points, $p = 0.060$), indicating that the COVID-era contribution (0.48 points) does not exceed pre-GFC levels (0.66 points). When we re-estimate using pre-GFC as the reference period, the COVID coefficient is 0.03 points and statistically indistinguishable from zero ($p = 0.816$). The government did not become a larger growth driver during COVID than it was before the GFC.

The descriptive context is essential for interpreting these magnitudes. Figure 1 shows mean government contribution by period with 95% confidence intervals. The government contribution declined dramatically from pre-GFC (0.66 pp) to post-GFC (0.23 pp)—the austerity trough—and partially recovered to 0.48 pp during COVID. Crucially, COVID-era levels remain below pre-GFC norms, and country-level dispersion (shown as overlaid points) is substantial within every period.

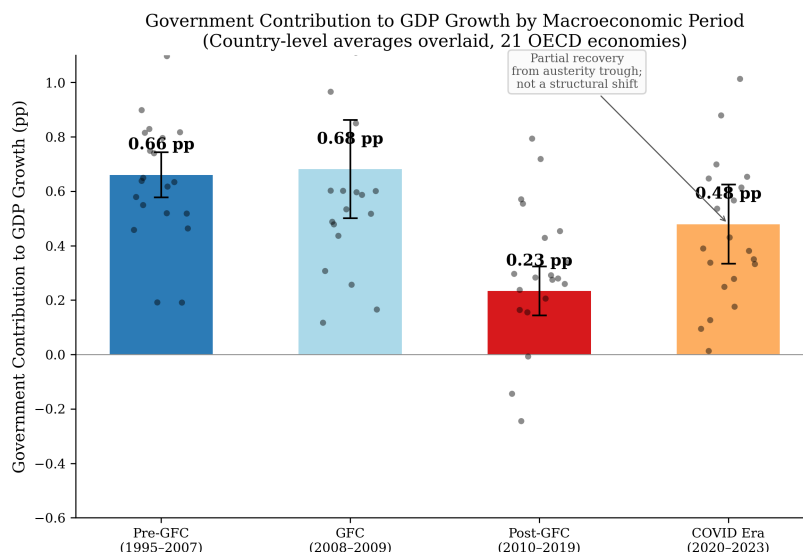


Figure 1: Government contribution to GDP growth by macroeconomic period. Bars show period means with 95% confidence intervals. Country-level averages are overlaid as scatter points, illustrating substantial cross-country dispersion within each period. COVID-era levels (0.48 pp) represent a partial recovery from the post-GFC austerity trough (0.23 pp) but remain below pre-GFC norms (0.66 pp).

Notes: Government contribution computed as the sum of real government consumption growth and real government GFCF growth, each weighted by lagged nominal GDP shares. Period definitions: pre-GFC (1995–2007), GFC transition (2008–2009), post-GFC (2010–2019), COVID-era (2020–2023). 21 OECD economies. Source: OECD National Accounts, World Bank WDI, authors' calculations.

Table 1 presents the baseline fixed effects estimates. The COVID-era coefficient (+0.25 pp, $p = 0.003$) is statistically significant but economically modest. Notably, the GFC transition period (+0.28 pp, $p = 0.012$) shows a comparable magnitude—both crisis episodes produced temporary government contribution increases of similar size, and neither represented a structural regime shift.

Figure 2 visualizes these period-dummy coefficients with 95% confidence intervals. The key insight is that the COVID-era effect, while statistically significant, is comparable in magnitude to both the pre-GFC period and the GFC transition—neither of which represented structural transformations. The COVID-era effect is a crisis response of familiar magnitude, not a regime shift.

Table 1: Baseline Fixed Effects Estimates: Government Contribution to GDP Growth

Variable	Coefficient	SE	p-value
Pre-GFC (1995–2007)	0.221	0.118	0.060
GFC transition (2008–2009)	0.282	0.112	0.012
COVID-era (2020–2023)	0.253	0.086	0.003
Country FE	Yes (21 countries)		
Controls	Yes (9 covariates)		
Reference period	Post-GFC (2010–2019)		
Observations	563		
R^2 (within)	0.293		

Notes: Dependent variable is government

contribution to real GDP growth in percentage points. Standard errors clustered at the country level (21 clusters). Controls include log GDP per capita, trade openness, population growth, public debt-to-GDP, unemployment, central bank policy rate, real effective exchange rate, household debt-to-GDP, and old-age dependency ratio. Country fixed effects absorb all time-invariant country-level heterogeneity. Reference period is post-GFC (2010–2019).

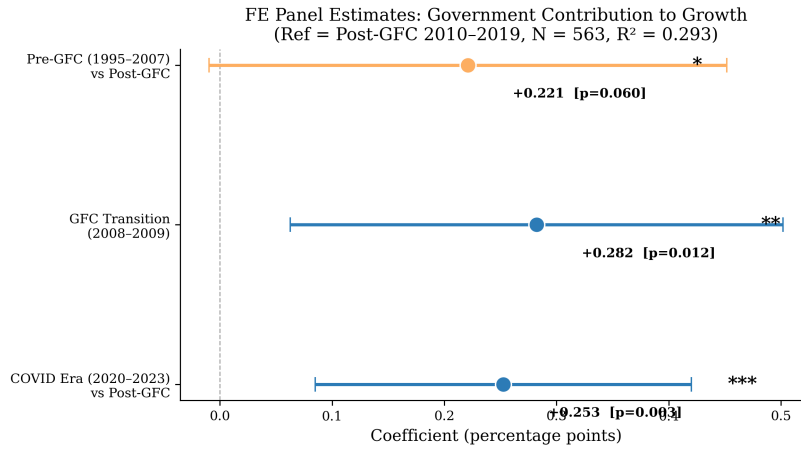


Figure 2: Fixed effects coefficient estimates with 95% confidence intervals. Reference period is post-GFC (2010–2019). The COVID-era coefficient (+0.25 pp) is statistically significant ($p = 0.003$) but comparable in magnitude to the pre-GFC period (+0.22 pp, $p = 0.060$) and GFC transition (+0.28 pp, $p = 0.012$). This confirms that COVID-era fiscal expansion represents a crisis response of familiar scale, not a structural regime shift.

Notes: Coefficients from baseline FE model (Model 1). Dependent variable is government contribution to real GDP growth (pp). Standard errors clustered at country level. Dashed line at zero represents the post-GFC reference period mean. 563 observations, 21 countries.

5.2 Durability: The Critical Test

The durability test delivers our most important finding. When we redefine the COVID era as 2021–2023, excluding the acute pandemic year, the coefficient drops to 0.13 points and loses statistical significance entirely ($p = 0.407$). The entire COVID-era government contribution increase is concentrated in 2020. By 2021–2023, the government’s contribution to growth had already normalized toward post-GFC levels. This is the decisive evidence against the structural transformation hypothesis: the historic fiscal expansion of 2020 did not produce a durable increase in the government’s role as a growth driver.

Figure 3 compares the COVID-era coefficient across all five specifications. The visual contrast between the significant baseline estimate and the non-significant R1 estimate (excluding 2020) provides the clearest summary of our central finding.

5.3 Growth Model Heterogeneity

Growth model heterogeneity is present but fragile. In Model 2, consumption-led economies show the largest COVID-era increase (0.39 points, versus 0.20 points for export-led economies), consistent with the hypothesis that economies reliant on private consumption had the greatest room for government-led demand substitution. The consumption-led

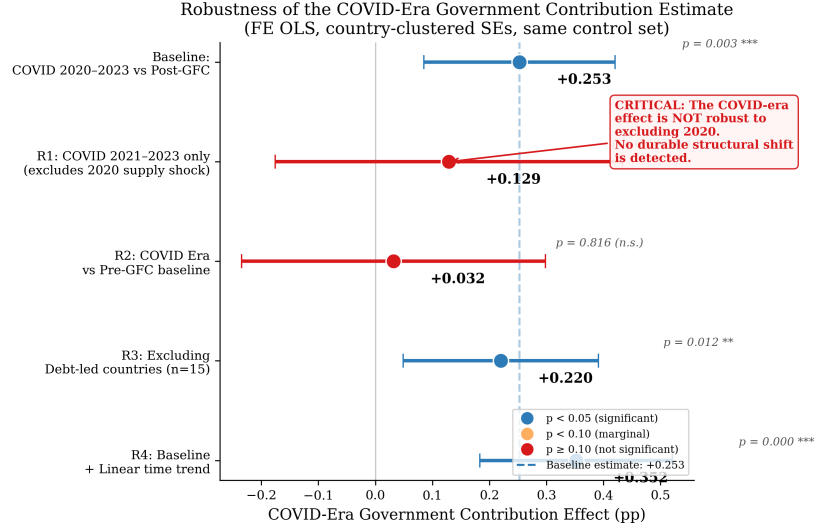


Figure 3: Robustness of the COVID-era coefficient estimate across five specifications. R1 (excluding 2020) is the decisive durability test: the coefficient drops to +0.13 pp and loses significance ($p = 0.407$). R2 (pre-GFC reference) shows COVID-era is not distinguishable from pre-GFC levels. R3 (excluding debt-led countries) confirms robustness in the non-debt-led subsample. R4 (linear time trend) amplifies the effect. The contrast between the significant baseline and the non-significant R1 estimate captures the paper’s central null finding.

Notes: Dots represent point estimates; horizontal lines represent 95% confidence intervals. All specifications estimated via country FE OLS with clustered standard errors. R1 uses 2021–2023 as COVID era. R2 uses pre-GFC (1995–2007) as reference period. R3 excludes six debt-led economies. R4 adds a linear time trend.

differential is 0.18 points, marginally significant ($p = 0.097$). Debt-led economies show no differential relative to export-led economies (-0.003 points, $p = 0.984$). However, when we control for the credit-to-GDP gap in Model 4, the consumption-led differential collapses to 0.03 points ($p = 0.778$). Growth model type is not independently informative about COVID-era fiscal patterns once financial structure is accounted for.

Figure 4 illustrates this finding. The left panel (Model 2) shows consumption-led economies with a substantially larger COVID-era effect, but the right panel (Model 4) shows this differential disappears when the credit-to-GDP gap is controlled. This is a substantively important null finding: financial cycle position, not growth model type, drives COVID-era spending heterogeneity.

5.4 Financial Cycle Amplification: The Strongest Finding

The credit-to-GDP gap is our strongest and most robust predictor of COVID-era government spending heterogeneity. Table 2 reports the full interaction results.

In normal times, a 10-percentage-point higher credit gap is associated with a 0.05-point *lower* government contribution ($p < 0.001$)—consistent with private credit crowding out government demand. During COVID, this relationship reverses: a 10-point higher pre-pandemic gap predicts an additional 0.17-point government contribution increase ($p < 0.001$), for a net COVID-era elasticity of +0.012 points per 10 points of gap. This result is robust in the full specification (Model 4: COVID $\times$ credit gap = 0.017, $p = 0.002$) and survives all robustness checks. A country at the 75th percentile of the credit gap distribution (+12 points above trend) experienced roughly 0.20 additional percentage points of government-led growth during COVID relative to a country at trend—a substantively meaningful amplification.

Figure 5 is the central visualization of this finding. The post-GFC fitted line shows the standard crowding-out pattern: higher credit gaps are associated with lower government contributions. The COVID-era fitted line shows the reversal: higher pre-pandemic credit gaps predict *larger* government contributions, consistent with government spending serving as the residual growth driver when private credit channels are exhausted.

The amplification logic is consistent with the Stockhammer, Wildauer, and Kohler (2020) framework: when private credit channels are exhausted or constrained—as they were in high-credit-gap economies entering the pandemic—

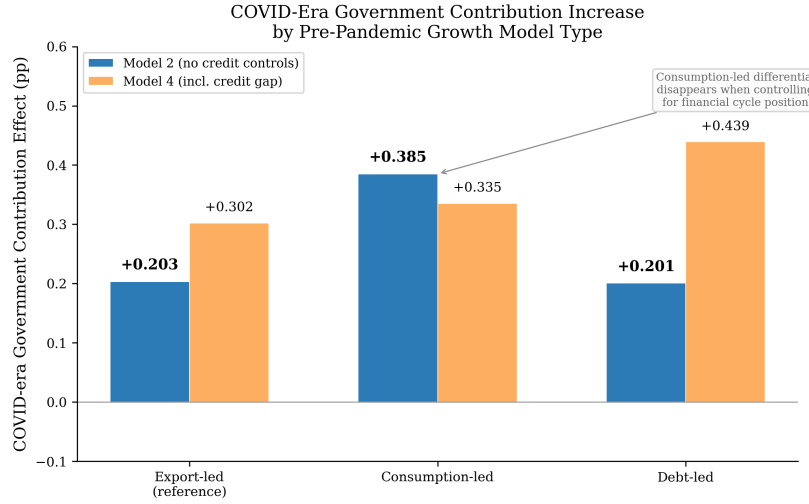


Figure 4: COVID-era government contribution effects by pre-pandemic growth model type. Left panel (Model 2): without credit-to-GDP gap controls, consumption-led economies show a larger COVID-era increase. Right panel (Model 4, full model): controlling for the credit-to-GDP gap, the consumption-led differential collapses to near zero. Financial cycle position, not growth model type, explains cross-country variation in COVID-era government spending patterns.

Notes: Bars represent total COVID-era effect (reference coefficient plus interaction differential). Export-led economies serve as the reference category. Model 2 includes period dummies and $\text{COVID} \times \text{growth model type}$ interactions. Model 4 additionally includes the credit-to-GDP gap and its interaction with the COVID dummy. Standard errors clustered at country level.

Table 2: Financial Cycle and Growth Model Heterogeneity

Variable	Model 2: GM Heterogeneity		Model 3: Financial Cycle		
	Coef.	<i>p</i>	Coef.	<i>p</i>	
COVID-era (export-led ref.)	0.203	0.022	0.340	<0.001	Notes: Dependent
COVID $\times$ Consumption-led	0.182	0.097	—	—	
COVID $\times$ Debt-led	−0.003	0.984	—	—	
Credit-to-GDP gap	—	—	−0.005	<0.001	
COVID $\times$ Credit gap	—	—	0.017	<0.001	
Country FE	Yes		Yes		
Controls	Yes		Yes		
Observations	563		563		

variable is government contribution to real GDP growth (pp). Standard errors clustered at the country level.

Credit-to-GDP gap is the deviation of private non-financial sector credit-to-GDP from its HP-filtered trend ($\lambda = 400,000$). Model 2 adds $\text{COVID} \times \text{growth model type}$ interactions. Model 3 adds the credit-to-GDP gap and its COVID interaction. Both models include the full set of controls and period dummies. Reference period is post-GFC (2010–2019).

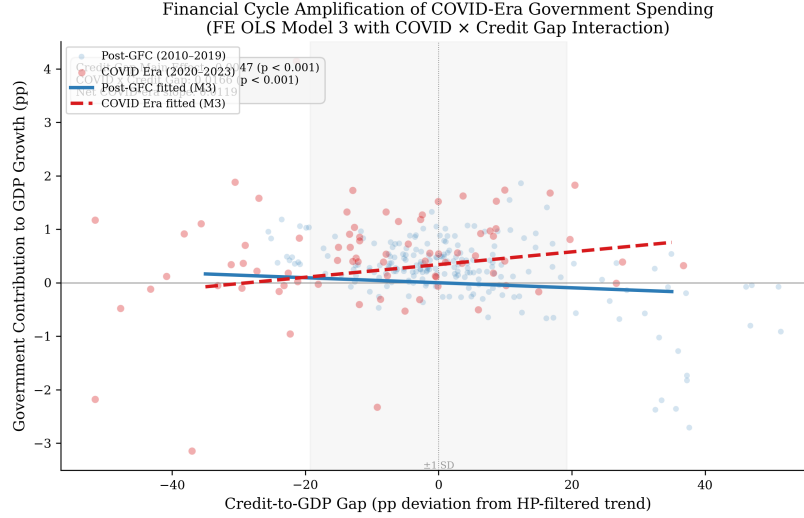


Figure 5: Credit-to-GDP gap interaction with government contribution. Scatter points represent country-year observations. Post-GFC fitted line (blue) shows the normal-times negative association: higher credit gaps predict lower government contributions (credit crowds out government). COVID-era fitted line (red) shows the reversal: higher pre-pandemic credit gaps predict *larger* government contributions ($p < 0.001$). This amplification is the strongest and most robust finding of the analysis, directly addressing Nölle (2023)’s call to integrate financial cycle dynamics into growth model analysis.

Notes: Credit-to-GDP gap is the deviation of private non-financial sector credit-to-GDP from its HP-filtered trend ($\lambda = 400,000$), following Borio and Drehmann (2009). Post-GFC observations: 2010–2019. COVID-era observations: 2020–2023. Fitted lines from bivariate linear regressions for visual clarity; formal estimation with country FE and controls reported in Table 2. BIS credit statistics, authors’ calculations.

government spending serves as the residual growth driver. But the durability evidence equally qualifies this interpretation: the amplification operated through the 2020 crisis response and did not persist into 2021–2023.

5.5 Robustness Summary

Our robustness checks confirm the central finding. Excluding debt-led countries leaves the baseline COVID coefficient stable at 0.22 points ($p = 0.012$), ruling out the concern that Southern European crisis dynamics drive the result. Adding a linear time trend strengthens the COVID coefficient to 0.35 points ($p < 0.001$), suggesting that the secular trend in government contribution has been slightly downward, making the COVID-era deviation larger relative to trend than relative to the post-GFC intercept.

In summary, we find a moderate, statistically significant increase in government contribution to growth during the COVID era. This increase falls far short of the hypothesized threshold for structural regime change. It is driven entirely by 2020 and does not survive exclusion of that year. The strongest predictor of cross-country variation in COVID-era fiscal expansion is not growth model type but financial cycle position—a finding that directly addresses the gap identified by Nölle (2023) and suggests that integrating financial cycle dynamics into the growth models framework is more important than expanding its typological categories.

6 Conclusion

The COVID-19 pandemic triggered the largest peacetime fiscal expansion in history. We asked whether this expansion represented a durable shift toward state-led growth across 21 OECD economies. The answer is no. The government contribution to GDP growth increased by 0.25 percentage points—four times smaller than the threshold for structural regime change—and the entire increase was concentrated in 2020. By 2021, the government’s growth contribution had already reverted toward post-GFC norms. The state-led growth model, while compelling as a theoretical possibility and championed by prominent economists (Mazzucato, 2013; Stiglitz, 2021), does not appear in the aggregate data for advanced economies through 2023.

Our strongest positive finding concerns financial cycles, not growth models. Countries with elevated pre-pandemic credit-to-GDP gaps deployed substantially more government spending during COVID, reversing the normal-times negative association between private credit expansion and government demand contributions. This finding directly fills the gap that Nölle (2023) identified—financial cycle dynamics matter for understanding government growth contributions, especially during crises—and suggests that the growth models framework would benefit more from integrating financial structure than from adding typological categories.

Three limitations define the boundaries of these conclusions. First, the COVID-era observation window is short. Four years of data cannot definitively distinguish a temporary crisis response from a slow-moving structural transformation. The durability verdict could change with data through 2025–2027. Second, our government contribution measure aggregates welfare-state spending with developmental-state investment. A qualitative shift in the composition of government spending—toward innovation policy, industrial strategy, and public investment—would not be fully captured by our aggregate measure. Third, our sample is restricted to advanced OECD economies. Emerging markets, where state intervention has different institutional meanings and the growth model taxonomy may not travel, could exhibit different dynamics.

Future work should extend the temporal coverage as new data become available, disaggregate government spending by function to distinguish welfare from developmental expenditure, and expand the sample to include emerging economies. The financial cycle amplification finding—that pre-pandemic credit conditions shape crisis fiscal response patterns—opens a broader research agenda on how financial structure and growth models interact, rather than treating them as parallel or competing explanatory frameworks.

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